

Xingfan Xia

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WORK EXPERIENCE

Compute Labs, Remote, Redmond, WA

- Co-Founder and Chief Technology Officer Aug 2024 – Present
 - **Leading the technological vision for democratizing AI compute access through blockchain-based tokenization, building the financial ecosystem for compute as an emerging asset class.**
 - Architecting the Compute Tokenization Protocol (CTP) that transforms physical GPUs (H100s, B200s) into tradeable digital assets, enabling fractional ownership and enhanced liquidity in AI infrastructure investments.
 - Directing R&D teams in developing the AI-Fi ecosystem that merges advanced AI infrastructure with DeFi protocols, creating innovative financial instruments and structured products for AI compute investments.
 - Spearheading the development of GNFT & COMPUTE token systems, implementing multi-faceted validation and benchmarking frameworks to ensure transparent, verifiable, and secure GPU tokenization transactions.
 - Building scalable blockchain infrastructure to optimize GPU utilization across decentralized and centralized compute networks, maximizing returns for investors throughout the AGI development journey.

Amazon Web Service – Athena, Redmond, WA

- Senior Software Engineer - AI/ML Infrastructure Apr 2022 – Present
 - **Building next-generation data infrastructure optimized for AI/ML workloads, enabling large-scale machine learning analytics on petabyte-scale datasets with billions of rows.**
 - **Tech Stack: Java, Python, Apache Spark, Apache Iceberg, AWS ML Services, Distributed Computing, Big Data Analytics**
 - Architected and implemented CTAS (Create Table As Select) feature for Iceberg Tables, enabling efficient data preprocessing for ML pipelines. Released at AWS RE:INVENT 2022 with millions of ML-focused queries executed since launch.
 - Optimized Hive JSON Serde library for ML data ingestion, improving feature extraction performance by 30% and ensuring deterministic results for model training datasets, reducing ML pipeline failures by 60%.
 - Developed a virtualized table connector enabling seamless access to diverse data formats for ML feature stores, improving data accessibility for AI applications by 40% and projected to save \$4M annually in ML infrastructure costs.
 - Led cross-team initiative to optimize query routing for AI/ML analytics workloads, reducing latency for real-time inference data queries and decreasing system congestion by 32%.

Apple, Cupertino, CA

- Machine Learning Engineer - Siri Intelligence Aug 2020 – Apr 2022
 - **Developing advanced AI systems and machine learning models to enhance Siri's intelligence across iOS, macOS, watchOS, and tvOS platforms.**
 - **Tech Stack: Python, Objective-C, TensorFlow, PyTorch, MLOps, Distributed ML, Apache Spark, Real-time ML Inference**
 - Designed and implemented on-device machine learning framework for user profiling, utilizing deep learning algorithms to predict user preferences and behavior patterns, serving as core intelligence layer for Apple Maps and Apple News recommendation systems.
 - Built automated MLOps pipeline for continuous model training and deployment, implementing A/B testing framework for ML experiments with automated model performance monitoring and rollback capabilities, reducing manual ML engineering overhead by 40%.
 - Developed contextual AI prediction service using ensemble learning methods to anticipate user intent based on temporal and spatial signals, powering intelligent suggestions in Siri Shortcuts and proactive recommendations.

Airbnb, San Francisco, CA

- Machine Learning Engineer - Trust & Safety AI Jan 2019 – Aug 2020
 - **Building AI-powered fraud detection and trust & safety systems using advanced machine learning techniques and real-time inference engines.**
 - **Tech Stack: Python, Java, Scala, TensorFlow, Scikit-learn, XGBoost, MLflow, Apache Kafka, Real-time ML Serving**
 - Developed sophisticated ML algorithms for fake review detection using NLP and anomaly detection techniques, reducing fraudulent reviews to 1% of peak volume and preventing \$2M in losses. Optimized model inference pipeline, improving detection speed by 30%.
 - Implemented ensemble learning approach for comprehensive user risk scoring system, combining multiple ML models to evaluate host and guest trustworthiness, similar to credit scoring systems but tailored for sharing economy platforms.
 - Deployed ML-driven policy engine for automated risk-based payout delays, using predictive models to identify high-risk transactions, preventing 60% of fraudulent payouts and generating \$10M in annual savings.
 - Architected real-time ML serving infrastructure using microservices, providing low-latency risk signals to downstream applications through Kafka streams and Redis caching for sub-100ms inference.

- Built scalable ML feature store and model serving platform supporting multiple fraud detection models, enabling rapid experimentation and deployment of new AI-powered safety features.

EDUCATION

Carleton College, Northfield, Minnesota

- B.A. in Computer Science

Graduated Jun 2018

TECHNICAL SKILLS

- Programming Languages: Proficiency in Python, Java, Golang, Javascript, Scala, Objective-C
- Experienced AWS Services: Athena, S3, SQS, EC2, Lambda, API Gateway, Auto Scaling Group, Cloudfront, Route 53, CodePipeline, CodeBuild, CodeDeploy, Kinesis
- Data Engineering & Analysis: Data Analysis Libraries, OpenCV, Visualization Tools, Scipy, Numpy, Pandas, ELK, Apache Iceberg, Apache Spark, Apache Kafka, Airflow